IS1212 : Probability and Statistics

K S Saubhagya
Lecturer (Probationary)
Department of Statistics, Faculty of Science, University of Colombo
saubhagya@stat.cmb.ac.lk

Independent Events

- Events A and B are said to be independent if one of the events can occur without being affected by the other.
- That is if A is independent of B, then the probability of the occurrence of A is unchanged by the information as to whether or not B has occurred.

- That is two events A and B are independent if;

$$P(A \cap B) = P(A)\,P(B)$$

Property If $P(A)>0$ and $P(B)>0$, then A and B are independent if and only if (iff)

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B)$$

Property

A and B are independent events implies that A and B′ are independent events

Independence of more than two events

- We illustrate the generalisation of the concept of independence to more than two events with reference to independence of three events.

Definition:

- Three events A, B and C are mutually independent if and only if

$$P(A \cap B) = P(A)\,P(B),$$

$$P(A \cap C) = P(A)\,P(C),$$

$$P(B \cap C) = P(B)\,P(C) \text{ and}$$

$$P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$$

Property

- If A, B and C are mutually independent events, then
  a) A and B $\cup$ C are independent events
  b) A and B $\cap$ C are independent events
  c) A, B and C' are mutually independent events
  d) A, B' and C' are mutually independent events etc.

Example

- A die is thrown twice. Find the probability of obtaining a 4 on the first throw and an odd number on the second throw. Check whether the A and B are independent.

Let A: "4 is obtained on the first throw". Then $P(A) = 1/6$

let B: "an odd number is obtained on the second die". Then $P(B) = 3/6$
$=1/2$

The result on the second throw is not affected in any way by the results on the first throw. Therefore, A and B are independent events.

And also, $P(A \cap B) = 3/36$ (show!!!)

That is $P(A \cap B) = 3/36 = 1/6*1/2 = P(A)*P(B)$

A and B are independent.

Examples…

- A bag contains 5 red counters and 7 black counters. A counter is drawn from the bag, the colour is noted and the counter is replaced. And a second counter is drawn. Let A be the event that the first counter is red, and B be the event that the second counter is black. Are the events A and B independent? Find $P(A \cap B)$.

Solution…

Let A: "the first counter is red". Then P(A)= 5/12

Let B: "the second counter is black"

since the first counter is replaced before the second draw is made, A and B are independent events.

Now, P(B)= 7/12

and

$$P(A \cap B) = P(A)P(B) = \left(\frac{5}{12}\right)\left(\frac{7}{12}\right) = \left(\frac{35}{144}\right)$$

Therefore A and B are independent.

Examples…

- Events A and B are such that $P(A)=1/3$, $P(A\cap B)=1/12$. If A and B are independent events, find

  a). $P(B)$ b). $P(A\cup B)$

Solution…

a). Since A and B are independent $P(A \cap B)=P(A)*P(B)=1/12$

$\Rightarrow P(B)= 1/4$

b).

$$P(A \cup B) = P(A)+P(B)-P(A \cap B)$$

$$= \frac{1}{3}+\frac{1}{4}-\frac{1}{12}=\frac{1}{2}$$

Examples…

- A fair die is thrown twice. Find the probability that
  a) Neither throw results 4
  b) At least one throw results 4

Solution…

a) A: obtain 4 in the first throw

B: obtain 4 in the second throw

$$P(\text{nether throw is } 4) = P(\overline{A} \cap \overline{B})$$

$$= P(\overline{A}) \times P(\overline{B}) = \left(\frac{5}{6}\right) \times \left(\frac{5}{6}\right) = \frac{25}{36}$$

b)

$$P(\text{at least one throw is } 4)$$

$$= 1 - P(\text{neither result is } 4) = 1 - \frac{25}{36} = \frac{11}{36}$$

Examples…

- Two events A and B are such that $P(A)=1/4$, $P(A|B)=1/2$, $P(B|A)=2/3$

a) Are the events A and B independent?
b) Find $P(A \cap B)$
c) Are A and B mutually exclusive?
d) $P(B)$

Solution…

a). If A and B are independent then P(A|B)=P(A). Since P(A|B)= ½ and P(A)=1/4, A and B are not independent.

b).

$$P(A \cap B) = P(B \mid A)P(A) = \left(\frac{2}{3}\right) \times \left(\frac{1}{4}\right) = \frac{1}{6}$$

c). If A and B are mutually exclusive then P(A ∩ B)=0. Since P(A ∩ B) =1/6 ≠ 0 A and B are not mutually exclusive events.

d).

$$\begin{aligned} P(A \mid B)P(B) &= P(B \mid A)P(A) \\ \left(\frac{1}{2}\right)P(B) &= \left(\frac{2}{3}\right)\left(\frac{1}{4}\right) \\ \Rightarrow \quad P(B) &= \frac{1}{3} \end{aligned}$$

Exercises

1) Toss a fair coin three times and observe the sequence of Heads and Tails. Define the following events

H1 = Heads on first toss H2 =Heads on second

toss H3 =Heads on third toss

A =all tosses the same

a) Are the events H1, H2, H3 mutually independent ?

b) Are the events A, H1 and H2 mutually independent?

Any questions?